CSE370:Database Systems

Lecture 7: Relational Algebra



BRAC UNIVERSITY

Inspiring Excellence

BACK TO SQL

How does a computer understand SQL queries? 🤔

Student

SID	Name	Dept	Attn_Rate
123	Meena	CSE	1.0
211	Raju	CSE	0.6
124	Mithu	CS	0.9
153	Rita	EEE	0.8
243	Geeta	ECE	0.75

```
-- QUERY
SELECT * FROM STUDENT
WHERE Dept='CSE' AND Attn_Rate > 0.7;
```

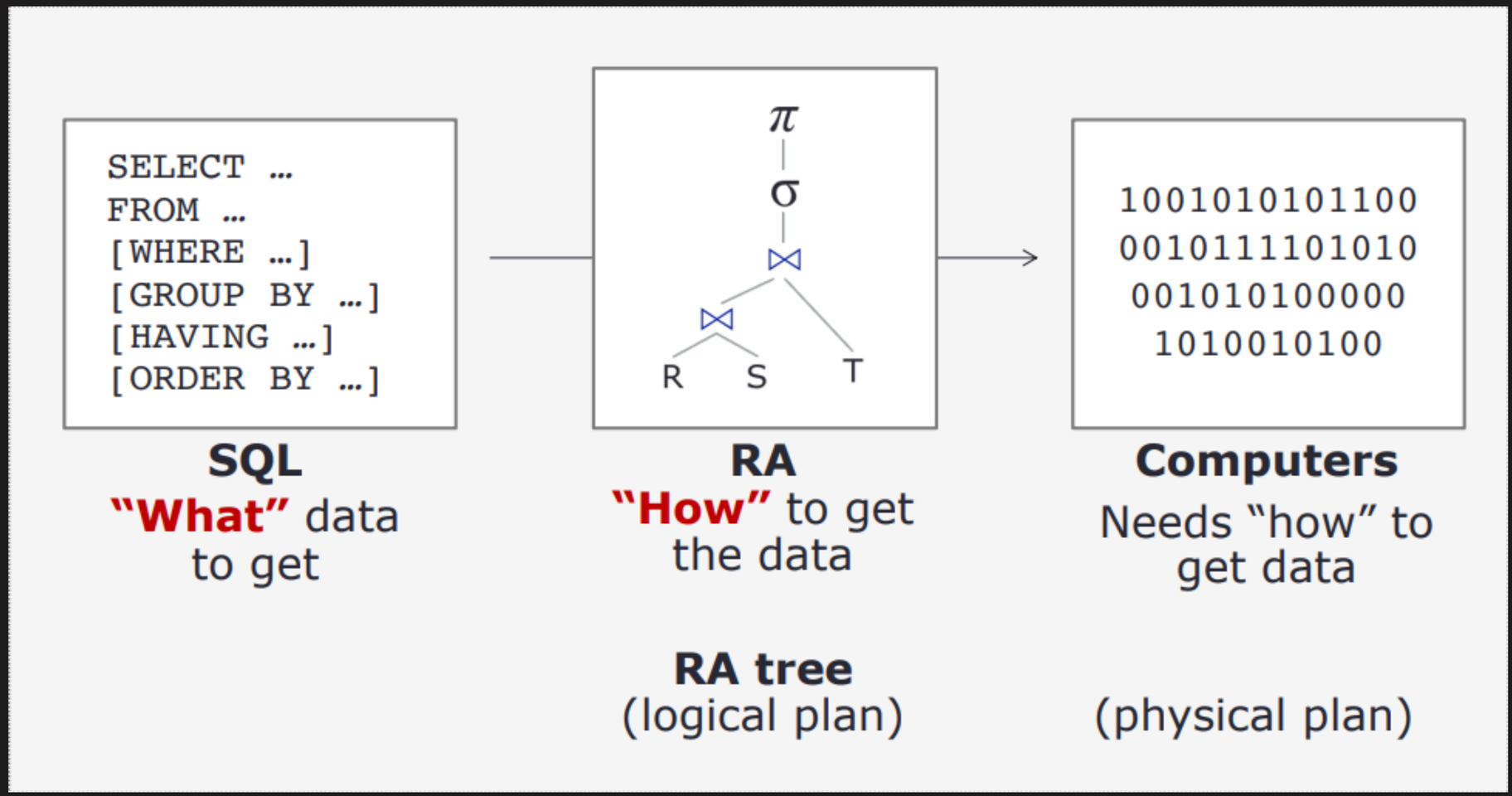


SID	Name	Dept	Attn_Rate
123	Meena	CSE	1.0

- When a RDBMS processes a query, it translates it into a RA tree
- After some optimization, the RA tree is converted into instructions

DBMS AND QUERY PROCESSORS

RDBMS



RELATIONAL ALGEBRA (RA)

Overview

- The basic set of operations for the relational model
- Enable a user to specify basic retrieval requests (or queries)
- The result of an algebra operation is a *new relation*

Unary Relational Operations

Take one relation, return a new relation

- SELECT (symbol: σ (sigma))
- PROJECT (symbol: π (pi))
- RENAME (symbol: ρ (rho))

Additional Expressiveness

Assignment (symbol: ←)

Aggregate Functions (symbol: G)

Binary Relational Operations

Take two relation, return a new relation

- Union (∪)
- Intersection (∩)
- Difference (−)
- Cartesian Product (×)
- Join (⋈ (theta))

SELECTION (σ)

$\sigma_c(R)$ - where C is a set of conditions that involve attribute(s) of R

- Unary operation – take one operand
- Return all tuples that satisfy a condition (filter tuples)
- Conditions can be =, <, ≤, >, ≥, ≠ and combined with AND, OR, NOT

Find students from CSE dept with attn_rate above 0.7

Student

SID	Name	Dept	Attn_Rate
123	Meena	CSE	1.0
211	Raju	CSE	0.6
124	Mithu	CS	0.9
153	Rita	EEE	0.8
243	Geeta	ECE	0.75

```
-- EQUIVALENT QUERY
SELECT * FROM STUDENT
WHERE Dept='CSE' AND Attn_Rate > 0.7;
```

$\sigma_{Dept='CSE'\ AND\ Attn_rate > 0.7}$ (Student)

SID	Name	Dept	Attn_Rate
123	Meena	CSE	1.0

PROJECTION (π)

$\pi_{A1,A2,....An}(R)$ - where A_i is attribute of a relation R

- Unary operation – take one operand
- Return specified attributes of a relation
- NO duplicates - Distinct values

Find all SID, Dept from Student

Student

SID	Name	Dept	Attn_Rate
123	Meena	CSE	1.0
211	Raju	CSE	0.6
124	Mithu	CS	0.9
153	Rita	EEE	0.8
243	Geeta	ECE	0.75

-- EQUIVALENT QUERY
SELECT DISTINCT SID, Dept FROM Student;

$\pi_{SID,\ Dept}$ (Student)

SID	Dept
123	CSE
211	CSE
124	CS
153	EEE
243	ECE

Solve π_{Dept} ($\pi_{SID,\ Dept}$ (Student))

Note: Projection will return a set i.e: no duplicate tuple will exist

RENAMING (ρ)

$\rho_{R'(B1, B2, \ldots, Bn)}(R)$ - where B_i is new attribute name of a relation R'

- Unary operation – take one operand
- Change the schema, not the instance
- Rename the result of expression E to R'
- Rename the resultant attributes to $B_1, B_2, \ldots, B_n$

Other Forms:

- **$\rho_S(R)$ - Changes relation name of result to S**
- **$\rho_{(b1, b2, \ldots, bn)}(R)$ - Changes attribute names of resultant**

Dept

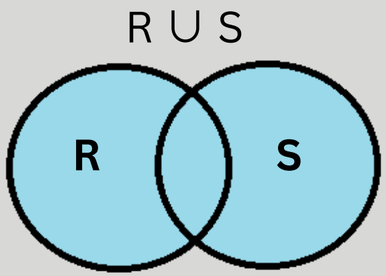
Initial	Name
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering

$\rho_{Departments(shortName, deptName)}(Dept)$

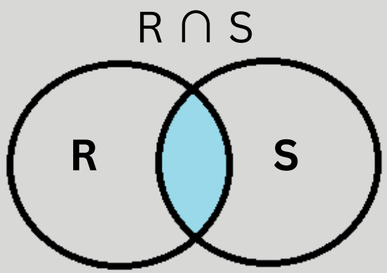
Departments

ShortName	DeptName
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering

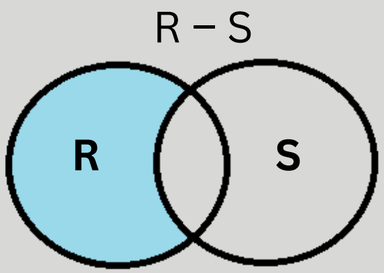
UNION, INTERSECTION, DIFFERENCE



Union



Intersection

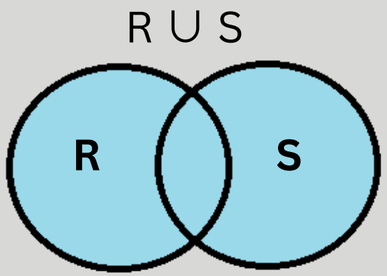


Difference

- Binary operation – takes two operand
- Both relations must have the same number of attributes and in same order
- Corresponding attributes of R and S must be based on the same domain/ compatible data types

UNION (U)

← Assignment operator



Union

Each tuple of R and each tuple of S is put into the resultant relation

Old_Depts

Initial	Name
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering

New_Depts

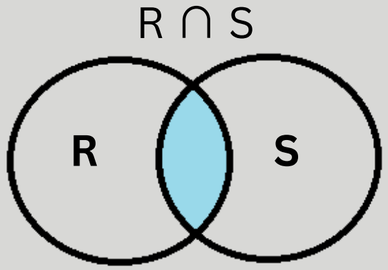
Initial	Name
AI	Artificial Intelligence
DS	Data Science
CS	Computer Science

All_Depts ← Old_Depts ∪ New_Depts

All_Depts

Initial	Name
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering
AI	Artificial Intelligence
DS	Data Science

INTERSECTION (∩)



Intersection

A tuple is in the resultant relation if and only if it is in both R and S

Old_Depts

Initial	Name
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering

New_Depts

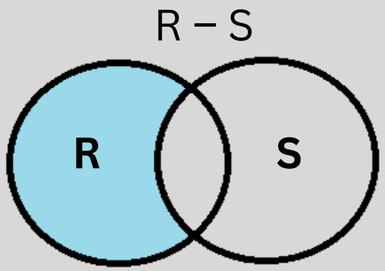
Initial	Name
AI	Artificial Intelligence
DS	Data Science
CS	Computer Science

Common_Depts ← Old_Depts ∩ New_Depts

Common_Depts

Initial	Name
CS	Computer Science

DIFFERENCE (–)



Difference

A tuple is in the resultant relation if and only if it is in R but not in S

Old_Depts

Initial	Name
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering

New_Depts

Initial	Name
AI	Artificial Intelligence
DS	Data Science
CS	Computer Science

Depts ← Old_Depts – New_Depts

Depts

Initial	Name
CSE	Computer Science and Engineering
EEE	Electrical and Electronics Engineering

Solve and compare with previous RA

Depts ← New_Depts – Old_Depts

CARTESIAN PRODUCT (×)

R x S - where $R(a_1, a_2, ..., a_n)$ and $S(b_1, b_2, ..., b_k)$

- Binary operation – takes two operand
- So-called “cross-product” or “product”
- Combine two relations
- Usually not meaningful when it is performed alone

$R \times S \neq S \times R$

StudentNames

ID	Name
123	Meena
234	Raju

StudentDepts

SID	Dept
123	CSE
234	CS

StudentNames x StudentDepts

ID	Name	SID	Dept
123	Meena	123	CSE
123	Meena	234	CS
234	Raju	123	CSE
234	Raju	234	CS

JOIN (⋈)

$R \bowtie_c S$ - where R and S has a set of attributes that are in common

- Binary operation – takes two operand
- Merge relations on the specified condition

$$R \bowtie_c S \equiv \sigma_c (R \times S)$$

StudentNames

ID	Name
123	Meena
234	Raju

StudentDepts

SID	Dept
123	CSE
234	CS

StudentNames $\bowtie_{ID = SID}$ StudentDepts

ID	Name	SID	Dept
123	Meena	123	CSE
234	Raju	234	CS

QUERY TREE (RA TREE)

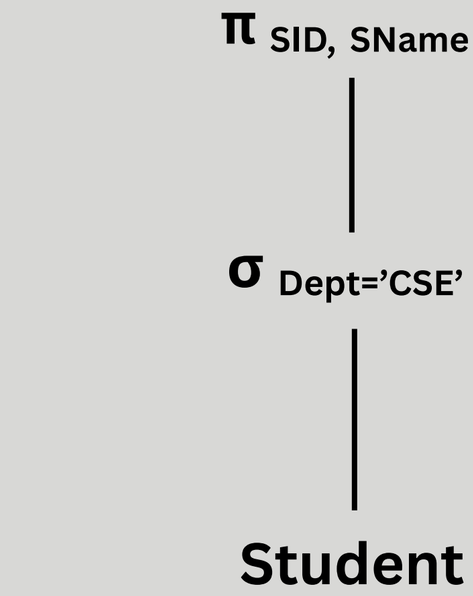
Overview

- An internal data structure to represent a query
- Estimates work involved in executing a query
- Nodes stand for ***operations (σ, π, ρ, ⋈)***
- Leaf Nodes represent ***base relations (tables)***
- Trees can be rearranged for better performance

Example:

Query Tree for the given RA:

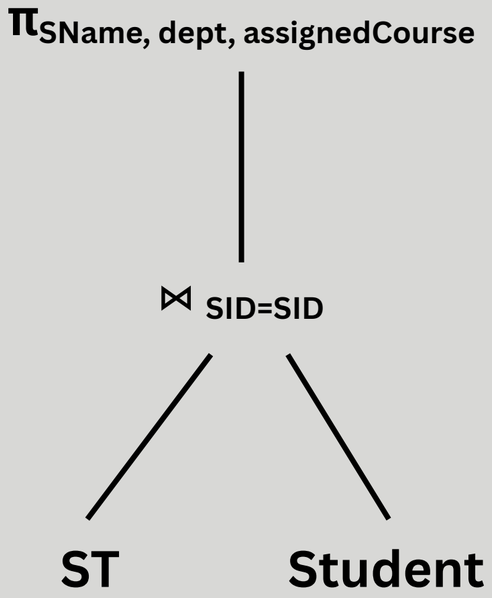
$\pi_{SID,\ SName}\ (\sigma_{Dept='CSE'}\ (Student))$



QUERY TREE (RA TREE)

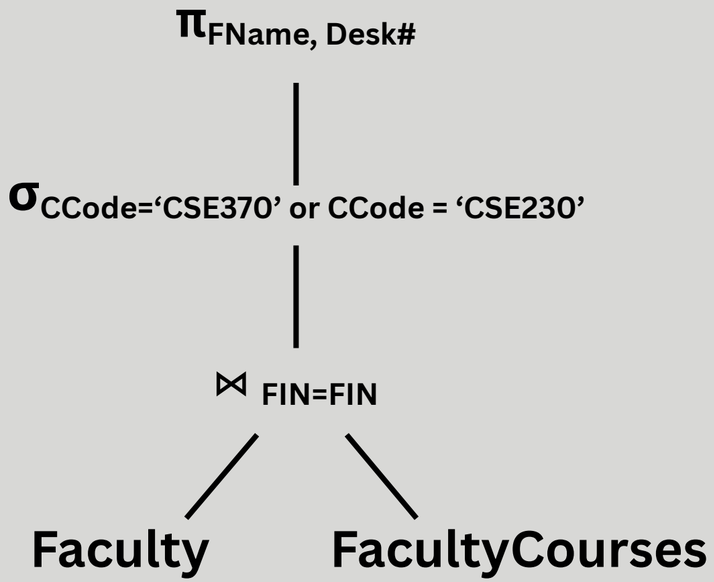
Query Tree for the given RA:

$\pi_{SName,\ dept,\ assignedCourse}(ST \bowtie_{SID=SID} Student)$



Query Tree for the given RA:

$\pi_{FName,\ Desk\#}(\sigma_{CCode='CSE370' \text{ or } CCode = 'CSE230'}(Facutly \bowtie_{FIN = FIN} FacultyCourses))$



Now Draw the RA Tree for the following RA:

$\pi_{FName,\ Desk\#}(\sigma_{CCode='Database\ Systems' \text{ or } CCode = 'AI'}((Facutly \bowtie_{FIN = FIN} FacultyCourses) \bowtie_{CCode\ =\ CCode} (Course)))$

Example Problems - I

Consider the following Relational Schema:

Student (SID, SName, address, dept, totalCredits)
ST(SID, assignedCourse)
Faculty(FIN, FName, LName, Desk#)
Course(CCode, CName)
FacultyCourses(FIN, CCode, Section)
-- FIN references Faculty(FIN), CCode references Course(CCode)

A) Write RA for the following:

1. Find students with at least 70 credits completed from ‘CS’ dept.
2. Find FName and Desk# of all faculties.
3. Find SID, SName of Students from ‘CSE’ dept.

Example - Solution (1)

A) Write RA for the following:

1. Find students with at least 70 credits completed from ‘CS’ dept.

Ans: $\sigma_{totalCredits>=70\ AND\ Dept='CS'}$ (Student)

2. Find FName and Desk# of all faculties.

Ans: $\pi_{FName,\ Desk\#}$ (Faculty)

3. Find SID, SName of Students from ‘CSE’ dept.

Ans: $\pi_{SID,\ SName}$ ($\sigma_{Dept='CSE'}$ (Student))

Example Problems - II

Consider the following Relational Schema:

Student (SID, SName, address, dept, totalCredits)
ST(SID, assignedCourse) -- *SID references Student(SID)*
Faculty(FIN, FName, Desk#)
Course(CCode, CName)
FacultyCourses(FIN, CCode, Section)
-- FIN references Faculty(FIN), CCode references Course(CCode)

B) Write RA for the following:

1. Rename CCode, CName columns of Course table to Code, CTitle
2. Find ST's SName, dept, assignedCourse.
3. Find Faculties FName and Desk# for those who are taking either CSE370 or CSE230 Courses.

Example - Solution (2)

B) Write RA for the following:

1. Rename CCode, CName columns of Course table to Code, CTitle

Ans: $\rho_{Code,\ CTitle}(Course)$

2. Find ST's SName, dept, assignedCourse.

Ans: $\pi_{SName,\ dept,\ assignedCourse}(ST \bowtie_{SID=SID} Student)$

3. Find Faculties FName and Desk# for those who are taking either CSE370 or CSE230 Courses.

Ans: $\pi_{FName,\ Desk\#}(\sigma_{CCode='CSE370' \text{ or } CCode = 'CSE230'}(Facutly \bowtie_{FIN = FIN} FacultyCourses))$

4. Find Faculties FName, Desk# for those who are taking either 'Database Systems' or 'AI' Courses.

Ans: $R \leftarrow ((Facutly \bowtie_{FIN = FIN} FacultyCourses) \bowtie_{CCode\ =\ CCode} (Course))$

$\pi_{FName,\ Desk\#}(\sigma_{CCode='Database\ Systems' \text{ or } CCode = 'AI'}(R))$

Note: ← (Assignment operator) can be used to decompose complex queries

What Next?

LECTURE 8: INDEXING



LOADING.....

